Data Analytics Internship – Task 1

Sales Performance Analysis

Submitted to: Alfido Tech

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Domain: Data Analytics Internship
Tools Used: Python, Pandas, Matplotlib, Google Colab

Submission Date: March 2026

1. Introduction

Data analytics is an important process that helps organizations understand patterns, trends, and insights from large amounts of data. By analyzing data effectively, businesses can make better decisions, improve their strategies, and enhance customer satisfaction. In today's digital world, companies rely on data-driven insights to understand customer behavior, identify market trends, and improve overall business performance.

The objective of this task is to analyze a sales dataset to understand customer purchasing behavior and sales performance across different categories and regions. The dataset contains information about orders, customers, product categories, and sales values. By analyzing these variables, it becomes possible to identify important patterns that can help businesses improve their marketing strategies and product offerings.

The analysis was performed using Python in Google Colab. Various exploratory data analysis techniques and visualizations were used to identify trends and patterns in the dataset.

2. Dataset Overview

The dataset contains sales transaction data including information about orders, customers, products, and regions. Important columns include Order ID, Customer Name, Category, Region, and Sales.

Customer_Behavior_Analysis.ipynb

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```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.rcParams["figure.figsize"] = (10,5)
sns.set_style("whitegrid")
```

[10] 32s

```python
from google.colab import files

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

df = pd.read_csv(file_name, encoding='latin1')

df.head()
```

Choose files superstore__taset (1).csv

superstore_final_dataset (1).csv(text/csv) - 2135456 bytes, last modified: 3/12/2026 - 100% done

Saving superstore_final_dataset (1).csv to superstore_final_dataset (1) (3).csv

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales
0	1.0	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.9600
1	2.0	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400
2	3.0	CA-2017-138688	12-06-2017	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.6200
3	4.0	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775
4	5.0	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold N Roll Cart System	22.3680

Next steps: Generate code with df New interactive sheet

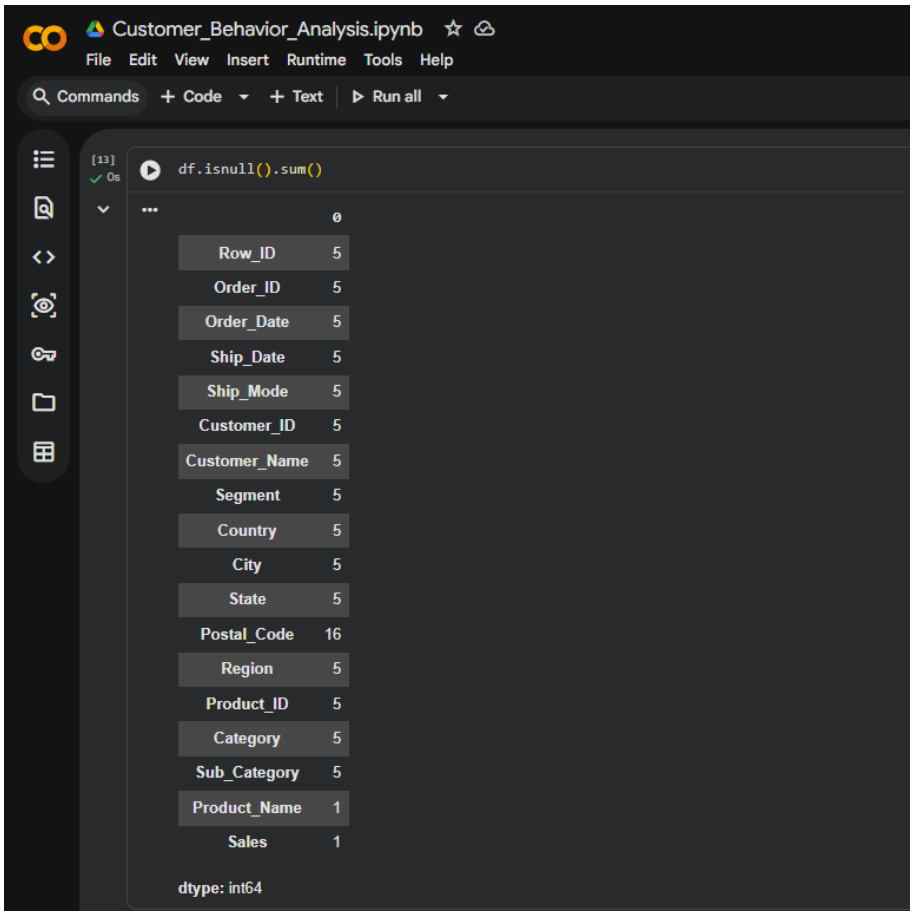
[11] 0s

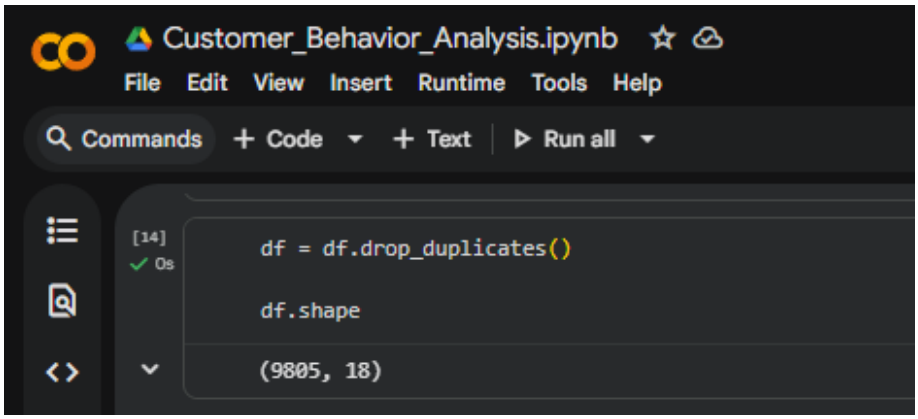
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9805 entries, 0 to 9804
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row_ID         9800 non-null   float64
 1   Order_ID       9800 non-null   object
 2   Order_Date     9800 non-null   object
 3   Ship_Date      9800 non-null   object
 4   Ship_Mode      9800 non-null   object
 5   Customer_ID    9800 non-null   object
 6   Customer_Name  9800 non-null   object
 7   Segment        9800 non-null   object
 8   Country        9800 non-null   object
 9   City           9800 non-null   object
 10  State          9800 non-null   object
 11  Postal_Code    9789 non-null   float64
 12  Region         9800 non-null   object
 13  Product_ID     9800 non-null   object
 14  Category       9800 non-null   object
 15  Sub_Category   9800 non-null   object
 16  Product_Name   9804 non-null   object
 17  Sales          9804 non-null   float64
dtypes: float64(3), object(15)
memory usage: 1.3+ MB
```

3. Data Cleaning

Data cleaning was performed to ensure data quality. Missing values were checked and duplicates were removed.





4. Summary Statistics

Summary statistics were calculated to understand the distribution of the dataset.

```
[12]
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df.describe()
```
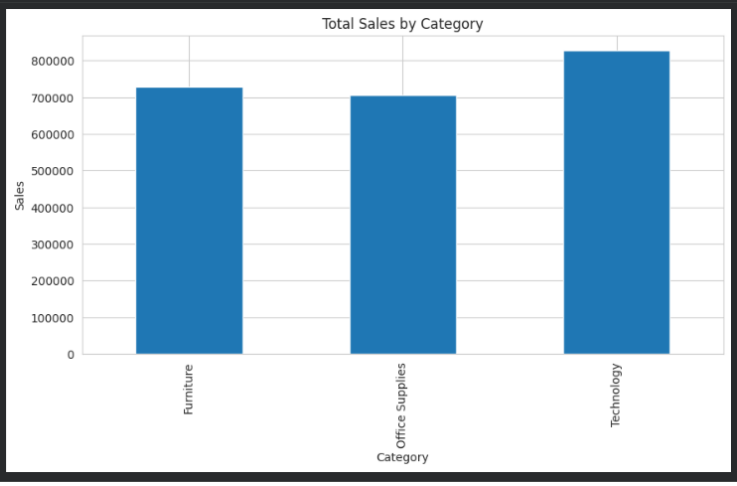
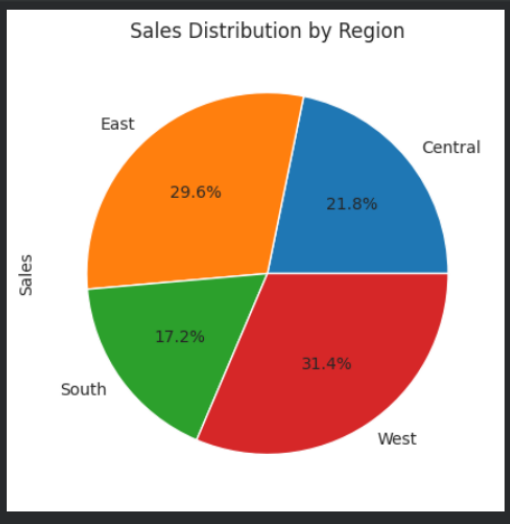

	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9804.000000
mean	4900.500000	55273.322403	230.705325
std	2829.160653	32041.223413	626.534737
min	1.000000	1040.000000	0.000000
25%	2450.750000	23223.000000	17.246000
50%	4900.500000	58103.000000	54.485000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

5. Data Visualization

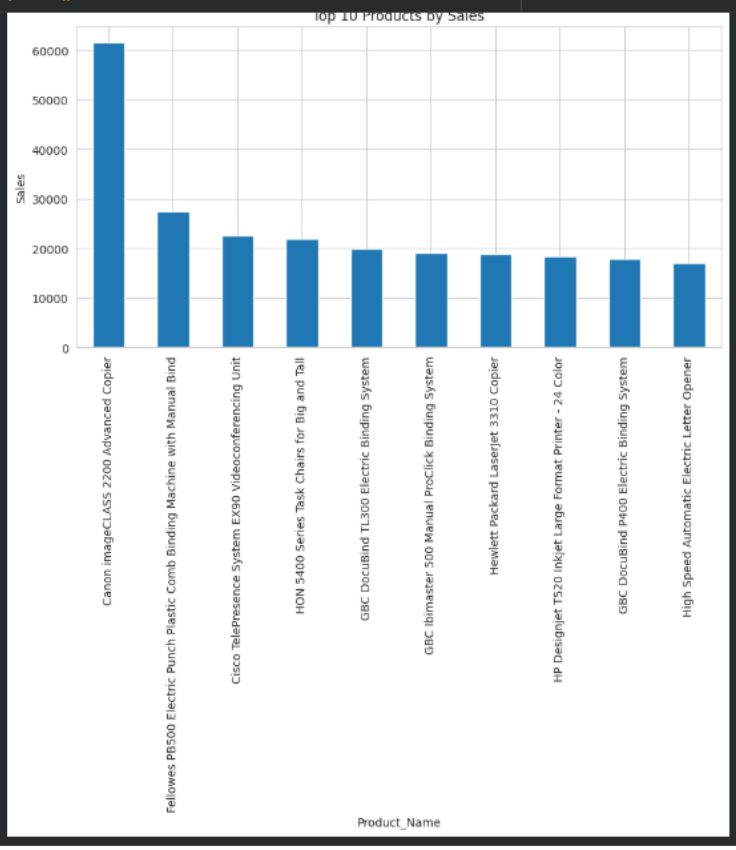
```
[15]
✓ 0s
category_sales = df.groupby("Category")["Sales"].sum()

category_sales.plot(kind="bar")

plt.title("Total Sales by Category")
plt.ylabel("Sales")
plt.show()
```



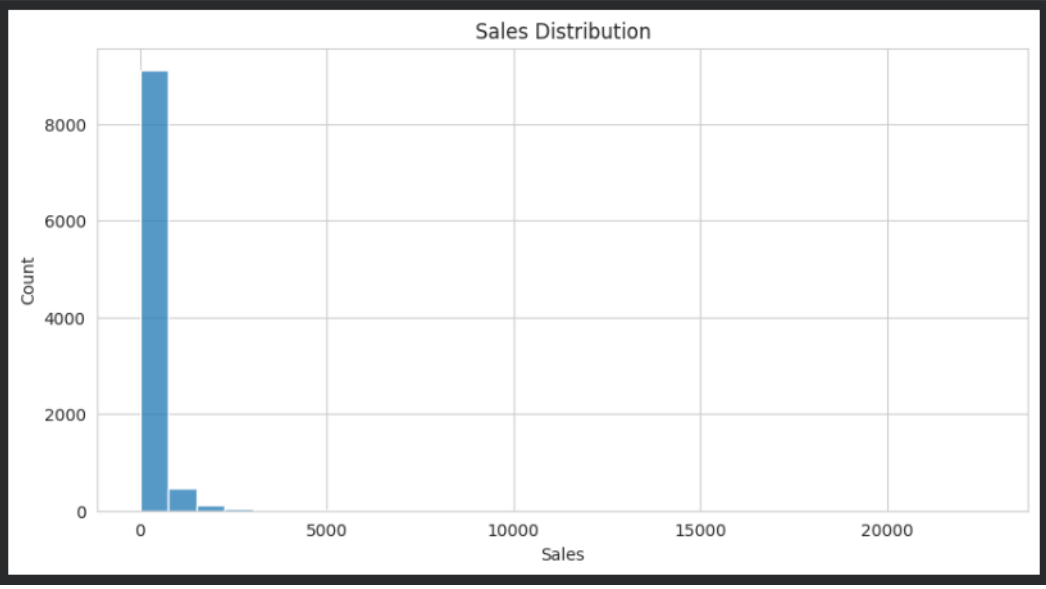
```
[16]
region_sales = df.groupby("Region")["Sales"].sum()

region_sales.plot(kind="pie", autopct="%1.1f%%")

plt.title("Sales Distribution by Region")
plt.show()
```




```
[20]
top_products = df.groupby("Product_Name")["Sales"].sum().sort_values(ascending=False).head(10)

top_products.plot(kind="bar")

plt.title("Top 10 Products by Sales")
plt.ylabel("Sales")
plt.show()
```



```
sns.histplot(df["Sales"], bins=30)

plt.title("Sales Distribution")
plt.show()
```



[21]
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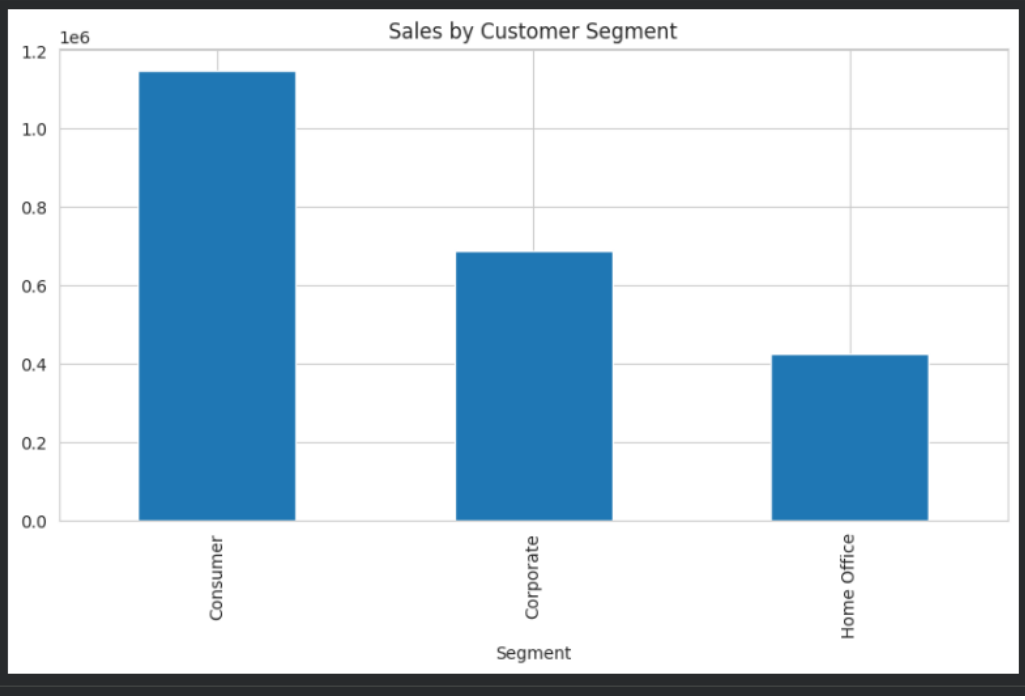
```
segment_sales = df.groupby("Segment")["Sales"].sum()

segment_sales.plot(kind="bar")

plt.title("Sales by Customer Segment")
plt.show()
```



6. Key Insights

1. Technology category generated highest sales.
2. Few products contribute most revenue.
3. Sales vary across regions.
4. Most sales are low-value.

7. Recommendations

1. Focus on high-performing categories.
2. Promote top products.
3. Improve low-performing regions.
4. Introduce loyalty programs.
5. Bundle products.

8. Executive Summary

This project analyzed sales data to identify patterns and trends. Visualizations helped understand category performance and regional distribution. Recommendations were provided to improve business strategies.

9. Project Links

https://github.com/Swayamchavan/alfido-data-analytics-internship